

REST API Documentation

Project Title: Library Management System (LMS)

Project Type: Full-stack Web Development Capstone Project

Base URL: http://localhost:8080

1. Books API – /api/books

Method	Endpoint	Description
GET	/api/books	Get all books
GET	/api/books/{id}	Get book by ID
GET	/api/books/isbn/{isbn}	Get book by ISBN
GET	/api/books/search/title	Search by title
GET	/api/books/search/author	Search by author
GET	/api/books/search/category	Search by category
GET	/api/books/search/publisher	Search by publisher
GET	/api/books/search/year	Search by exact year
GET	/api/books/search/year-range	Search by year range
POST	/api/books	Add new book
PUT	/api/books/{id}	Full update book
PATCH	/api/books/{id}/copies	Update copies
DELETE	/api/books/{id}	Delete book

Detailed Descriptions

GET /api/books

→ Success (200 OK): Array of Book objects

```
{
  "id": 5,
  "isbn": "978-0134685991",
  "title": "Effective Java",
  "author": "Joshua Bloch",
  "publisher": "Addison-Wesley",
  "publicationYear": 2018,
  "category": "Programming",
  "totalCopies": 10,
  "availableCopies": 7
}
```

GET /api/books/{id} and GET /api/books/isbn/{isbn}

- Success (200 OK): Same single Book object
- Error (404): { "message": "Book not found" }

POST /api/books

- Request Body:

```
{  
  "isbn": "978-0596009205",  
  "title": "Head First Design Patterns",  
  "author": "Eric Freeman",  
  "publisher": "O'Reilly",  
  "publicationYear": 2004,  
  "category": "Design Patterns",  
  "totalCopies": 8  
}
```

- Success (201 Created): Returns created Book with ID

PATCH /api/books/{id}/copies

- Request Body: { "total": 15, "available": 10 }
- Success (200 OK): Updated Book object

DELETE /api/books/{id}

- Success (204 No Content)
-

2. Authentication & Users API

Method	Endpoint	Description
POST	/api/users/login	Login
POST	/api/users	Register user
GET	/api/users/{id}	Get user profile
GET	/api/users/check-email	Check email

Detailed Descriptions

POST /api/users/login

- Request:

```
{  
  "email": "john@example.com",  
  "password": "secret123"  
}
```

- Success (200 OK):

```
{
  "id": 10,
  "email": "john@example.com",
  "fullName": "John Tan",
  "role": "Member"
}
```

→ Error (401): "Invalid email or password"

POST /api/users

→ Request Body:

```
{
  "email": "jane@example.com",
  "fullName": "Jane Lim",
  "password": "password123",
  "role": "Member"
}
```

→ Success (201 Created): Created user with ID

GET /api/users/{id}

→ Success (200 OK): Full user object (same as login response)

GET /api/users/check-email?email=test@example.com&excludId=10

→ Success (200 OK): { "available": true }

3. Members API

Method	Endpoint	Description
GET	/api/members/{userId}	Get member profile
PATCH	/api/members/{userId}/status	Update status
PATCH	/api/members/{userId}/renew	Renew membership

Detailed Descriptions

GET /api/members/{userId}

→ Success (200 OK):

```
{
  "memberId": "MEM0001",
  "status": "Active",
  "joinDate": "2024-01-15",
  "dueDate": "2025-01-15"
}
```

PATCH /api/members/{userId}/status

- Request Body: { "status": "Suspended" }
- Success (200 OK): Updated member object

PATCH /api/members/{userId}/renew

- No request body
 - Success (200 OK): Updated joinDate & dueDate
-

4. Borrowing API – /api/borrows

Method	Endpoint	Description
GET	/api/borrows/user/{userId}/active	Get active loans
GET	/api/borrows/user/{userId}/summary	Get borrow history
POST	/api/borrows	Borrow book
PATCH	/api/borrows/{id}/return	Return book
PATCH	/api/borrows/{id}/renew	Renew loan
PATCH	/api/borrows/{id}/pay-fine	Pay fine

Detailed Descriptions

GET /api/borrows/user/{userId}/active

- Success (200 OK): Array of active borrows
{
 "id": 101,
 "book": { "id": 5, "title": "Clean Code" },
 "borrowDate": "2025-03-20",
 "dueDate": "2025-04-03",
 "fineAmount": 2.50,
 "timesRenew": 1
}

POST /api/borrows

- Request: { "memberUserId": 10, "bookId": 5 }
- Success (201): Created borrow record

PATCH /api/borrows/{id}/return

- Success (200): Updated borrow with returned = true

PATCH /api/borrows/{id}/renew

- Success (200): Updated dueDate

PATCH /api/borrows/{id}/pay-fine

→ Success (200): fineAmount = 0.00

5. Reservations API

Method	Endpoint	Description
POST	/api/reservations	Reserve book
PATCH	/api/reservations/{id}/fulfill	Issue book
DELETE	/api/reservations/{id}	Cancel

Detailed Descriptions

POST /api/reservations

→ Request: { "memberUserId": 10, "bookId": 5 }
→ Success (201): Created reservation

PATCH /api/reservations/{id}/fulfill

→ Success (200): Creates borrow record

DELETE /api/reservations/{id}

→ Success (204 No Content)